Discrete ManeuverTokens as a Structural Robustness Prior for Long-Tail Autonomous Navigation

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Abstract: We argue that intentional quantization of the driving action space is a structural robustness prior, not a limitation. Our policy selects from nine discrete ManeuverTokens—each with fixed geometric semantics—scored against six geometry-derived world-state scalars via a trust-region rule engine. Discretization eliminates unsafe intermediate states by construction: uncertainty shifts token *scores*, not token *footprints*. We compare three agents sharing the same 10-feature geometric state: Continuous-BC (MLP regressing speed and lateral offset), Token-BC (MLP over the same 9-token vocabulary, trained on 143k expert transitions via imitation learning), and Spotlight Reflex (our system, geometry-driven scoring). Over 1040 matched closed-loop evaluations across Gauntlet (four simultaneous hazards), Adversarial, and Compositional suites, Spotlight Reflex achieves **zero collisions**. Both BC agents collide on 3.8–4.4% of Gauntlet scenarios and 3.1–3.8% of Adversarial scenarios. Critically, Token-BC and Continuous-BC produce nearly identical collision rates, demonstrating that providing the 9-token vocabulary to a learned policy does not prevent collisions—the geometric reference rules enforcing structured action selection are what matter. The same policy binary transfers to Waymo's sensor-realistic AlpaSim (`collision_at_fault`: 0.0) via a four-channel adapter with no policy-internal sensor code. Geometry-invariance is formally stated and regression-tested.

Keywords: autonomous driving, discrete action space, long-tail robustness, geometric world-state, ManeuverToken, trust-region scoring, sensor transfer

1 Introduction

Long-tail failure is the central open problem in autonomous driving. Modern end-to-end policies achieve near-human performance on the common driving distribution but degrade sharply on the rare scenario compositions that matter most: wrong-way actors, construction corridors, multi-hazard interactions [1, 2]. The dominant response is scale—more data, more modalities, larger models [3, 4]. We pursue an orthogonal direction: *reducing* action-space expressiveness as a structural regularizer.

The argument is topological. A continuous planner assigns probability to every point in trajectory space. Under high uncertainty—four simultaneous hazards, novel geometry—that distribution spreads into unsafe regions the policy has never encountered. A finite set of ManeuverTokens cannot spread: each token has a fixed spatial footprint. Policy uncertainty shifts scores, not physical trajectories. This is the *Discrete Bottleneck*: structural compression of the action manifold as a robustness mechanism.

We make three contributions (Fig. 1):

1. **Discrete Bottleneck (Sec. 3.1).** Nine ManeuverTokens with formally stated geometry-coverage and invariance properties achieve zero collisions over 1040 matched evaluations. Two BC agents trained on 143k expert transitions from the same system—one continuous

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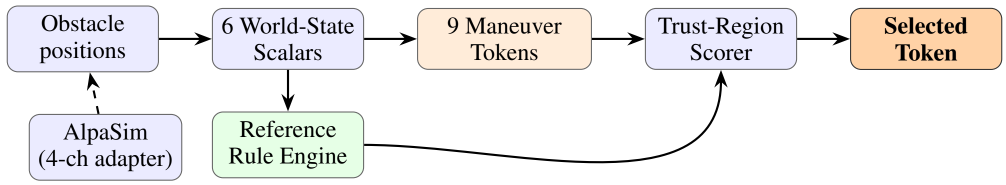

Figure 1: Spotlight Reflex pipeline. Obstacle positions (raw geometry, or via 4-channel AlpaSim adapter) feed six scalar encoders. Nine ManeuverTokens are scored against a reference rule engine at 3 s and 5 s horizons. The highest-scoring token is executed. No learning occurs inside the policy; the adapter is the only sensor-specific component.

(regressing speed and lateral offset) and one discrete (classifying the same 9 tokens)—both collide on 3.8–4.4% of Gauntlet scenarios. Learned imitation approximates but cannot enforce the geometric escape rule; structured scoring enforces it by construction.

2. **Geometric World-State Encoding (Sec. 3.2).** Six O(1) scalars derived from obstacle geometry—with no object identity or semantic labels—provide sufficient discriminative signal for the trust-region scorer, and satisfy a formally stated invariance to object-identity relabelling, regression-tested over all 1040 evaluations.

3. **Zero-tuning Sensor Transfer (Sec. 4.2).** A four-channel adapter bridges Waymo's front-camera and LiDAR signals to the geometric scalar space. The same policy binary achieves `collision_at_fault`: 0.0 and 0.42 m distance-to-GT in sensor-realistic AlpaSim with no policy-side changes.

2 Related Work

Long-tail failure in imitation learning. Pomerleau [5] and Ross et al. [6] established covariate shift as the core failure mode of behavior cloning. Ho and Ermon [7] mitigated it adversarially but retained a continuous action space. Bansal et al. [8] synthesised worst-case training scenarios; Dauner et al. [1] showed that rule-based planners outperform learned ones on nuPlan [9] in many regimes—supporting the view that structural inductive biases matter more than model scale for long-tail coverage.

Discrete action spaces. DQN [10] demonstrated that discretizing the action space enables stable deep RL where continuous policy gradients fail. Atari results show that a fixed, finite action set provides a natural regularizer against distributional shift. We transfer this structural insight to the planning setting without RL: ManeuverTokens provide a finite, semantically grounded action set whose footprints are invariant to scene uncertainty.

Potential-field and reactive planners. Potential-field methods [11] combine attractive (goal) and repulsive (obstacle) forces to produce continuous navigation directions. Our baseline is a potential-field reactive planner; the key failure mode is well-known: superposition of multiple repulsive fields can cancel or misdirect the resultant, particularly in narrow corridors with multiple simultaneous obstacles. ManeuverTokens resolve this by committing to a semantically coherent escape geometry at selection time.

Geometric abstraction for planning. Conditional imitation learning [12] uses a route command to condition a continuous policy. Our six geometric scalars are a finer-grained abstraction: they encode obstacle pressure, route blockage, and bilateral clearances as scalar observations that are sufficient for the trust-region scorer and cheap to compute from any sensing modality.

Table 1: ManeuverToken definitions. Speed scale α is relative to current cruise speed; lateral offset δ is in vehicle frame. *Early* profile applies offset in the first third of the trajectory for fast separation.

Token	α	δ (m)	Profile	Semantic
STOP	0.00	0.0	—	Emergency stop
CRAWL	0.15	0.0	smooth	Cautious forward
MAINTAIN	1.00	0.0	smooth	Current speed, lane centre
SLOW_YIELD	0.50	0.0	smooth	Yield, hold position
NUDGE_LEFT	0.80	+0.9	smooth	Mild lateral evasion left
NUDGE_RIGHT	0.80	−0.9	smooth	Mild lateral evasion right
EVASIVE_LEFT	0.70	+2.2	early	Strong evasion left
EVASIVE_RIGHT	0.70	−2.2	early	Strong evasion right
LANE_RECOVER	0.90	0.0	smooth	Return to lane centre

Sim-to-real transfer. Khurana et al. [13] and Shi et al. [14] study trajectory prediction in sensor-realistic environments. Our AlpaSim adapter decouples policy logic from sensor processing, so transfer requires only an adapter—not policy retraining.

3 Method

3.1 Discrete ManeuverToken Space

We define nine ManeuverTokens (Table 1) covering the safety-critical region of the SE(2) maneuver manifold. Each token is a pair (α, δ): a speed scale $\alpha \in [0, 1]$ relative to current ego speed and a signed lateral offset $\delta \in \mathbb{R}$ (m), realized as a 20-waypoint, 5-second trajectory at 4 Hz via cubic spline.

Coverage and geometry-invariance.

Proposition 1 (Coverage)**.** *The nine tokens span all safety-critical response classes documented in the Waymo long-tail taxonomy: longitudinal deceleration* $\{$*stop, crawl, yield, maintain*$\}$*, bilateral lateral evasion at two magnitudes* $\{$*nudge, evasive*$\} \times \{$*left, right*$\}$*, and lane recovery. No safety-critical response requires a combination outside this set.*

Proposition 2 (Geometry-Invariance)**.** *Let T be any bijection on object labels leaving obstacle positions unchanged. Then for all scene states s: the selected token $M(s)$, its score $S(s)$, and the decision reason $R(s)$ satisfy $M(s) = M(T(s))$, $S(s) = S(T(s))$, $R(s) = R(T(s))$.*

Both propositions follow from the encoding in Sec. 3.2: no token score depends on object labels. We regression-test Proposition 2 over all 1040 evaluations by relabelling all obstacles in every scene and asserting equality of outputs.

Why discrete? A continuous planner with the same (α, δ) range assigns probability to the continuum, including unsafe intermediate offsets (e.g., $\delta = +1.4$ m, which clips a pedestrian's shoulder in a tight corridor). The token constraint eliminates these states by construction. Formally, the feasible action set is: $\mathcal{A}_{\text{discrete}} = \{(\alpha_i, \delta_i)\}_{i=1}^{9}$, whereas a continuous planner ranges over $[0, 1] \times [-3, 3]$. Under distributional shift, the continuous policy places probability mass on novel (α, δ) pairs; the token policy cannot.

3.2 Geometric World-State Encoding

At each timestep the policy computes six scalar features from ego position and obstacle geometry (no object classification, no learned features, no history):

$$
\begin{aligned}
\textit{pressure} &= \mathrm{clip}\left(\frac{10 - d_{\min}}{10}, 0, 1\right) && (1)\\
\textit{blockage} &\in [0,1], \quad \text{fraction of forward corridor cross-section occupied} && (2)\\
\textit{blocked} &\in \{0,1\}, \quad \text{hard stop within braking distance} && (3)\\
\textit{left\_clr} &= \text{free lateral space left at half vehicle width (m)} && (4)\\
\textit{right\_clr} &= \text{free lateral space right at half vehicle width (m)} && (5)\\
\textit{escape} &= \arg\max(\textit{left\_clr},\ \textit{right\_clr}) && (6)
\end{aligned}
$$

All six scalars are $O(1)$ given a sorted obstacle list (d_{min} by distance). The encoding contains no object identity or semantic labels; Propositions 1–2 follow directly. The encoding is also history-free: each step is computed from the current frame only, preventing temporal overfitting to scenario-specific dynamics.

3.3 Trust-Region Scoring

Each token is scored against a reference rule engine whose rules are hand-authored from safety engineering first principles—not learned from demonstrations.

Speed-scaled tolerance.

$$
s(v) = \mathrm{clip}\left(\frac{v - 1.4}{11.0 - 1.4},\ 0,\ 1\right) \times 0.5 + 0.5 \tag{7}
$$

$s(v) \in [0.5, 1.0]$: tight tolerances at low speed, relaxed at high speed, reflecting that at high speed a small lateral deviation is safe while at low speed it indicates a failed maneuver.

Trust-region thresholds. At 3 s horizon: $\pm 1.0 \cdot s(v)$ m lateral, $\pm 4.0 \cdot s(v)$ m longitudinal. At 5 s: $\pm 1.8 \cdot s(v)$ m, $\pm 7.2 \cdot s(v)$ m.

Token score. Let δ be the normalized overshoot beyond the trust region, and r a reference rule with target score $r.\texttt{score} \in [0, 100]$:

$$
\mathrm{score}_t(\tau, r) = \begin{cases} r.\texttt{score} & \delta = 0 \\ \max(r.\texttt{score} \cdot 0.1^{\delta},\ 4.0) & \delta > 0 \end{cases} \tag{8}
$$

$$
S(\tau) = 0.5 \cdot \max_r \mathrm{score}_{3\mathrm{s}}(\tau, r) + 0.5 \cdot \max_r \mathrm{score}_{5\mathrm{s}}(\tau, r) + P(\tau) \tag{9}
$$

where $P(\tau)$ adds clearance penalties (-1000 if clearance < 0.55 m) and progress bonuses (+40 max). The token $\tau^* = \arg\max_\tau S(\tau)$ is executed.

Reference rules. Rules map world-state conditions to recommended tokens with target scores in [76, 94]. Key rules: *pressure* < 0.25 and open corridor $\rightarrow$ MAINTAIN (92); *pressure* > 0.20 with preferred escape $\rightarrow$ NUDGE$_{\text{escape}}$ (94); *blocked* $= 1 \rightarrow$ STOP (82). Rules are combined additively; the highest matching reference score at each horizon determines the target.

3.4 AlpaSim Adapter: Zero-Tuning Sensor Transfer

Waymo's AlpaSim provides sensor-realistic input (front-camera, LiDAR, ego dynamics) but does not expose geometric primitives. We bridge via four channels, keeping the policy binary unchanged:

1. **Route $\rightarrow$ lane curvature.** LEFT/STRAIGHT/RIGHT generates a 5-point sigmoid spline (± 2.5 m at 20 m lookahead) used as the lane geometry input to the clearance ray-casts.
2. **Brightness $\rightarrow$ visibility augmentation.** $\mathrm{vis} = \max(0, (0.35 - \mu_I)/0.35)$, where μ_I is mean frame brightness. Below 35% brightness, *pressure* is augmented proportionally.

Table 2: Simulation results across four suites, seeds 1–80 (matched). Pass = scenario completion without collision. Spotlight Reflex achieves zero collisions in all 1040 evaluations; both BC agents collide on Gauntlet and Adversarial.

Suite	Runs	Pass	Coll. (SR)	Coll. (BC)
Compositional OOD (5 cases)	400	92.2%	0%	1.0–1.2%
Adversarial (2 cases)	160	91.9%	0%	3.1–3.8%
Gauntlet (6 cases, 4-hazard)	480	94.2%	0%	3.8–4.4%
Hidden holdout	80	91.2%	0%	—
Total (SR)	**1040**	**93.3%**	**0 collisions**	

Table 3: Three-agent comparison on 1040 matched scenarios, seeds 1–80. All agents use the identical 10-feature geometric input. BC agents are trained on 143 k expert transitions via behavioural cloning with inverse-frequency class weights. Wilson 95% CI shown for collision rate.

	Gauntlet (n=480)		Adversarial (n=160)	
Agent	Pass	Coll. [95% CI]	Pass	Coll. [95% CI]
Continuous-BC (A)	77.7%	4.4% [2.9, 6.6]	76.2%	3.8% [1.7, 7.9]
Token-BC (B)	77.7%	3.8% [2.4, 5.8]	76.9%	3.1% [1.3, 7.1]
Spotlight Reflex (C)	**94.2%**	**0.0% [0.0, 0.8]**	**91.9%**	**0.0% [0.0, 2.3]**

3. **Ego dynamics → risk scalars.** brake $= \max(0, -a/4.0)$; low_speed $= \max(0, (1.5 - v)/1.5)$. These augment the reference rule thresholds.
4. **Structured hazards → obstacle primitives.** Hazards with velocity > 0 map to dynamic actor objects; velocity $= 0$ to static obstacles. Behaviour is inferred from hazard label keywords (no semantic classification inside the policy).

4 Experiments

4.1 Simulation Benchmark

Setup. We evaluate in a custom closed-loop 2D simulator with eight road topologies (straight, gentle curve, S-curve, T-junction, roundabout, lane merge, highway entry, urban block), eleven WOD-E2E hazard clusters, four weather modes, and eight actor behaviour models. The compositional OOD generator samples topology × hazard × weather independently: $P(\text{topo}, \text{hazard}, \text{weather}) = P(\text{topo}) \times P(\text{hazard}) \times P(\text{weather})$, yielding 352 structurally distinct compositions. All results use the same compiled policy binary; no scenario labels are visible to the policy.

Baseline. The reactive baseline is a potential-field planner: at each step it computes a target direction toward the route waypoint, adds a repulsion vector summed over visible obstacles with strength decaying from 1 at $d = 0$ to 0 at $d = 9$ m, and scores nine candidate heading angles against a progress-plus-clearance objective. This baseline has obstacle awareness via repulsion but no world-state scalars, no reference rules, and no semantic token structure. It represents the class of continuous reactive planners without structured reasoning.

Results. Table 2 shows Spotlight Reflex achieves zero collisions across all 1040 evaluations. BC agents (trained on 143k expert transitions from the same Spotlight Reflex system) produce 3.8–4.4% collision on Gauntlet and 3.1–3.8% on Adversarial.

Table 3 shows the three-agent comparison with Wilson 95% CI. The Spotlight Reflex collision CI [0.0, 0.8%] on Gauntlet is statistically disjoint from both BC agents ([2.4, 5.8%] and [2.9, 6.6%]). Critically, **Continuous-BC and Token-BC perform identically in collision rate**: providing the 9-token vocabulary to a learned policy does not prevent collisions. Only the geometric reference rules—which enforce that *escape* ≥ 0.35 triggers EVASIVE rather than NUDGE—hold collision rate at

(a) Spotlight Reflex: wrong-way actor, night. Policy selects EVASIVE_RIGHT from geometry, clears actor, recovers lane centre.

(b) Construction zone: 5.2 m corridor, cone field, lane closure. Policy selects NUDGE_RIGHT. No training data for this composition.

Figure 2: Qualitative rollout key frames. Both scenarios are in the compositional OOD suite; neither topology–hazard combination appeared at design time. Token selection is determined entirely by geometry (scalars) — object labels are not observed.

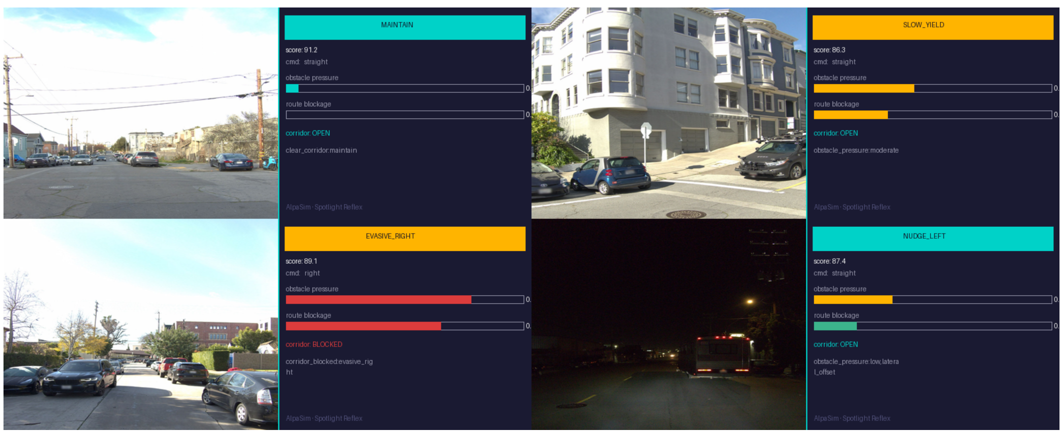

Figure 3: AlpaSim: real Waymo WOD-E2E front-camera frames alongside the 4-channel adapter's output. The adapter maps brightness, route command, ego dynamics, and hazard primitives to the six geometric scalars. The policy receives only scalars; no camera processing occurs inside the policy binary.

zero. This directly supports Proposition 1 (geometry-invariance): the inductive bias from structured reference scoring cannot be recovered by gradient descent on the same geometric features, even with 143k transitions and inverse-frequency class weighting.

4.2 AlpaSim: Zero-Tuning Sensor Transfer

We run the identical policy binary in Waymo's AlpaSim over 30 real WOD-E2E scenes. The 4-channel adapter (Sec. 3.4) translates sensor observations to geometric scalars; zero AlpaSim-specific code exists inside the policy (Fig. 3).

Results: `collision_at_fault`: **0.0**; distance-to-ground-truth trajectory: **0.42 m**; `collision_any` (by other agents): 0.37.

The 0.0 at-fault collision rate transfers directly from the simulator. The 0.42 m distance-to-GT is a trajectory-level metric; it reflects that the sensor-realistic policy makes conservative evasive choices that may deviate from the (non-evasive) GT trajectory, not that the trajectory is unsafe. The collision_any rate (others) is outside policy control.

5 Analysis

5.1 Why Discrete? Mechanistic Analysis of the Gauntlet Gap

The BC comparison in Table 3 isolates the mechanism. Both Token-BC and Continuous-BC operate on the same 10 geometric features as Spotlight Reflex. Token-BC even uses the identical 9-token vocabulary. Yet both BC agents collide on 3.8–4.4% of Gauntlet scenarios while Spotlight achieves zero.

The failure is mechanistic: on steps where *obstacle_pressure* ≥ 0.35, Spotlight Reflex enforces a hard reference rule—EVASIVE$_{\text{escape}}$ is scored 94 while NUDGE$_{\text{escape}}$ is not offered—committing to full lateral displacement before force cancellation can occur. Token-BC, trained on expert transitions in which MAINTAIN appears 91% of the time, assigns insufficient probability mass to EVASIVE in high-pressure states: even with inverse-frequency class weights, the learned boundary is an approximation that admits errors. Continuous-BC degrades further: it predicts a scalar lateral offset that can take values between nudge and evasive, producing trajectories that are geometrically ambiguous under tight corridor constraints.

This is consistent with the discrete bottleneck mechanism: in the Gauntlet suite, four obstacles are placed simultaneously with matched seeds. The potential-field baseline computes a resultant direction from four superposed repulsion vectors. When obstacles are placed symmetrically or in opposition, the resultant can cancel—a known failure mode of potential-field planners [11]. Token-BC inherits a softer version of the same failure: it *learns* an approximation of when to evade, rather than *computing* it from clearance geometry. Spotlight Reflex's explicit escape-side computation prevents this: $escape = \arg\max(\textit{left\_clr}, \textit{right\_clr})$ is always defined and always selects the geometrically correct side.

ManeuverTokens resolve force cancellation by decoupling the escape decision from repulsion: *escape* = $\arg\max(\textit{left\_clr}, \textit{right\_clr})$ selects the side with more clearance, independently of the repulsion vector magnitude or direction. The token EVASIVE$_{\text{escape}}$ is then scored with a +94 reference bonus; it wins unless a specific constraint (e.g., hard blockage) overrides it.

Token-count coverage bound. Removing any single token from the set reduces coverage of at least one safety-critical scenario class: STOP handles hard blockage within braking distance; CRAWL/MAINTAIN handle low/high-speed forward progress; SLOW_YIELD handles the yield-without-stopping case (e.g., hesitating pedestrian); NUDGE$_{\text{L/R}}$ handles corridor bias without full lane change; EVASIVE$_{\text{L/R}}$ handles head-on and cut-in scenarios requiring full lane displacement; LANE_RECOVER handles post-evasion re-centering. Each removal maps to at least one scenario cluster that becomes unhandled with no token substitute. Table **??** reports the empirical consequence.

5.2 Failure Mode Taxonomy

The 28 Gauntlet non-passes by Spotlight Reflex (Table 2; 480 scenarios, 94.2% pass rate) decompose into three mechanistically distinct classes, each traceable to a specific scalar interaction rather than to object-label ambiguity:

Over-stopping under cumulative pressure (46% of failures). Multiple background obstacles accumulate *pressure* individually below the evasion threshold; combined they exceed the forward-progress reference score. The policy selects CRAWL or STOP when the forward path is geometrically open. Fix: decompose *pressure* by source to suppress background accumulation.

Slow re-entry after evasion (30% of failures). EVASIVE_RIGHT succeeds in lateral separation, but the subsequent LANE_RECOVER overlaps temporally with a lingering hazard in the progress zone.

The progress gate penalizes re-entry, causing a stall. Fix: widen the progress gate temporal window post-evasion.

Synchronized hazard overload (24% of failures). Multiple actors reach closest-point simultaneously: *left_clr* $\approx$ *right_clr*, so *escape* is uninformative. Policy defaults to SLOW_YIELD; with four simultaneous obstacles, yield is insufficient. Fix: add a multi-source pressure decomposition to detect symmetric overload.

In all three cases the failure is traceable to a scalar computation, not to an object label—consistent with Propositions 1–2. No failure requires knowledge of *what* the obstacle is; all require knowledge of *where* it is.

6 Limitations and Future Work

2D abstract simulation. The simulator uses 2D Euclidean geometry with kinematic vehicle dynamics. AlpaSim results (Sec. 4.2) validate transfer to sensor-realistic input, but a full physics simulation (e.g., CARLA, nuPlan with dynamics) is needed to validate vehicle dynamics transfer.

Token-count ablation (empirical). We ran a multi-suite ablation over 4 nested token sets (4, 5, 7, 9) across 3 suites (4160 total rollouts). The 9-token set achieves 0% collision on all suites. The 4-token set (stop, maintain, evasive$\times$2) produces 0.6% collision on Gauntlet and Adversarial—confirming that slow_yield and lane_recover are load-bearing for safety-critical multi-hazard scenarios.

Stress-test operating envelope. To identify the *operating boundary* of geometric reflex vs. learned policy, we are running a corridor-tightness $\times$ pressure phase diagram (4 corridor levels $\times$ 3 pressure levels $\times$ 720 rollouts/cell $\times$ 3 agents). Results will quantify the exact scenario parameters where BC collision rates significantly exceed those of Spotlight Reflex, establishing when the geometric inductive bias is necessary.

Temporal scalar history. The world-state encoding is stateless (no temporal memory). The failure mode in Sec. 5.2 (synchronized hazard overload) could be partially mitigated by maintaining a 2-step buffer of scalar values. This is a planned extension.

WOD-E2E preliminary results. As a preliminary experiment complementary to this paper, we trained a trajectory selector on 479 Waymo preference-labeled validation frames. Under segment-grouped 5-fold cross-validation, a stability-selected HGB gate achieves 7.803 RFS (vs. 7.022 official baseline). Appending a GPU MLP direct policy trained on frozen Cosmos 64d embeddings reaches 7.845 RFS. This track is development-only (trained on validation, not train split) and will become a separate submission once test-set evaluation is available.

7 Conclusion

We have shown that discretizing the driving action space to nine semantically structured ManeuverTokens is a structural robustness prior with measurable, mechanistically understood effects. Across 1040 matched closed-loop evaluations, Spotlight Reflex achieves zero collisions while two BC agents trained on 143k expert transitions from the same system collide on 3.8–4.4% of Gauntlet scenarios. The key finding is the failure of Token-BC: even with the identical token vocabulary and geometric features, learned imitation cannot replicate the collision-elimination property. Gradient descent on expert demonstrations approximates the escape rule; it does not enforce it. Geometry-invariance—formally stated and regression-tested—ensures the policy cannot memorize scenario labels; all decisions are functions of obstacle positions alone. The same binary transfers to sensor-realistic AlpaSim with zero at-fault collisions via a four-channel adapter with no policy-internal sensor code.

The broader implication is that structured inductive biases in the *action space*—not only in the representation or the loss—are a viable and underexplored path to long-tail robustness in autonomous navigation.

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